

boilers 4 rows <i>one row = one boiler · the asset register</i> boiler_id PKinternal id, 1–4 codeplant tag 'B-2101' name'North watertube' maker_model'Babcock FM-120' rating_tphdesign output, t/h design_bargdesign pressure, barg commissionedyear in service fuel'Natural gas'	operators 6 rows <i>one row = one person on the roster</i> operator_id PK1 ... 6 emp_no'EM-4471' full_name'Somchai Srisuk' shift'A', 'B' or 'C' cert_class'Class 1' / 'Class 2' Values in 'quotes' are text — everything else is a number.	water_tests 45 rows <i>one row = one weekly water sample</i> test_id PK1 ... 45 boiler_id FK→ boilers.boiler_id test_date'YYYY-MM-DD' silica_ppmppm (plant limit 10) conductivity_usµS/cm phosphate_ppmresidual, ppm pHpH of the sample
readings 1,866 rows <i>one row = one instrument reading, every 4 h</i> reading_id PK1 ... 1866 boiler_id FK→ boilers.boiler_id ts'YYYY-MM-DD HH:MM' steam_tphsteam flow, t/h stack_temp_cflue gas, °C o2_pctflue gas O ₂ , %	daily_logs 311 rows <i>one row = one boiler on one day</i> log_id PK1 ... 311 boiler_id FK→ boilers.boiler_id operator_id FK→ operators.operator_id log_date'YYYY-MM-DD' steam_tsteam that day, t fuel_gjgas that day, GJ	maintenance_events 14 rows <i>one row = one job done on one boiler</i> event_id PK1 ... 14 boiler_id FK→ boilers.boiler_id event_date'YYYY-MM-DD' event_type'Tube cleaning', 'Burner service', 'Inspection', ... hours_downtime off line, h noteswhat the fitter wrote
fuel_prices 3 rows <i>one row = what gas cost in one month</i> month PK'2026-02' ... '2026-04' baht_per_gjgas price, ฿/GJ Join it on a month you cut out yourself: ON f.month = substr(d.log_date,1,7)	boiler_id is not the tag read this twice 1 = B-2101 North · 2 = B-2103 South 3 = B-2102 Centre · 4 = B-2104 New The big tables store only the id, so never name a machine from its number — JOIN boilers b ON b.boiler_id = r.boiler_id and read the tag off b.code.	clause order always this order WITH a named sub-query optional SELECT which columns you want FROM which table JOIN...ON how another table matches WHERE which rows to keep GROUP BY what to summarise by HAVING which groups to keep ORDER BY how to sort DESC = big first LIMIT how many rows to show Only SELECT and FROM are compulsory.

From the question to the query			
"How many ...?"	→ COUNT(*) · COUNT(DISTINCT x)	"The worst / the top five"	→ ORDER BY x DESC + LIMIT 5
"For each boiler / shift / day"	→ GROUP BY that column	"Show the tag, not the id"	→ JOIN the small table
"Only where ..."	→ WHERE rows · HAVING groups	"How often is it above X?"	→ WHERE x > X then COUNT(*)
"Per tonne / per hour / per day"	→ divide one SUM by another	"Compared with the average"	→ > (SELECT AVG(x) FROM t) in the WHERE
"Why did it change?"	→ look it up in maintenance_events	"Is it getting worse?"	→ GROUP BY substr(ts, 1, 7) — by month

Traps that catch everybody	The shape of a real query
- Text needs single quotes: code = 'B-2103', never "B-2103". Numbers take none. - WHERE filters rows, HAVING filters groups — WHERE AVG(x)>150 is an error. - Check MIN and MAX before you trust an AVG — a dead thermocouple reading 0.0 and a spike of 511.8 are both hiding in readings. - A rate is SUM(fuel_gj)/SUM(steam_t), not AVG(fuel_gj/steam_t) — the second lets a quiet day count as much as a busy one. - Anything in SELECT beside an aggregate must appear in GROUP BY too. - Put COUNT(*) next to every average — unequal counts mean an outage, a new machine, or missing data. - A join need not be on an id — ON f.month = substr(log_date,1,7) works.	<pre>-- is any boiler getting worse? SELECT substr(d.log_date, 1, 7) AS month, b.code, ROUND(SUM(d.fuel_gj)/SUM(d.steam_t), 3) AS gj_t, COUNT(*) AS n_days FROM daily_logs d JOIN boilers b ON b.boiler_id = d.boiler_id GROUP BY substr(d.log_date, 1, 7), b.code HAVING COUNT(*) > 20 ORDER BY b.code, month Pieces worth remembering: AVG MIN MAX SUM COUNT · ROUND(x, 1) · substr(ts, 1, 10) the day · substr(ts, 1, 7) the month · x IN ('B','C') · (SELECT AVG(x) FROM t) as one value</pre>